



Proton NMR Spectroscopy Video Series 2

Video 1:
NMR Spectra of Mixtures





Video Series Goals

Video 1: NMR Spectra of Mixtures (**Current**)

- Diagnostic NMR signals
- Calculating Percent Composition of a Mixture

Video 2: NMR Solvents

- Deuterated Solvents
- Solvent Signals in NMR

Video 3: Splitting Pattern Complexity

- Extension of “n+1” Rule

Video 4: Long Range Coupling

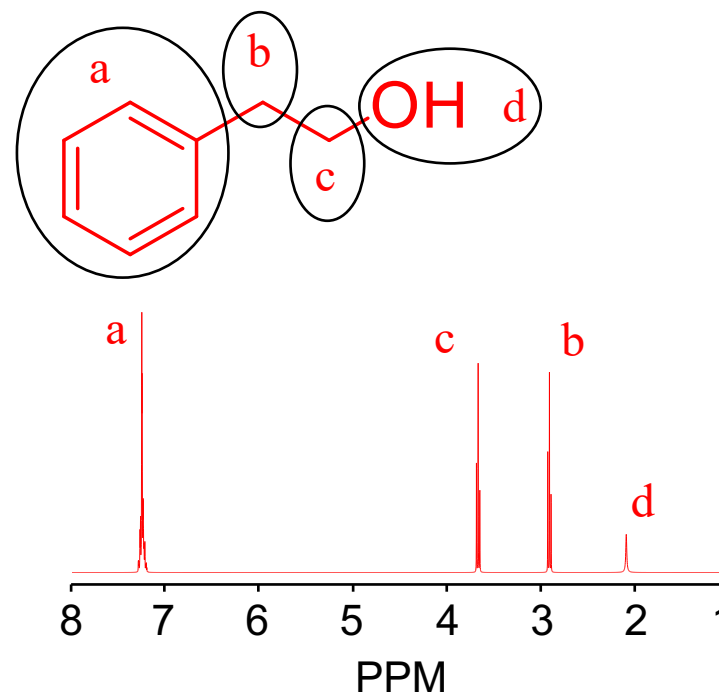
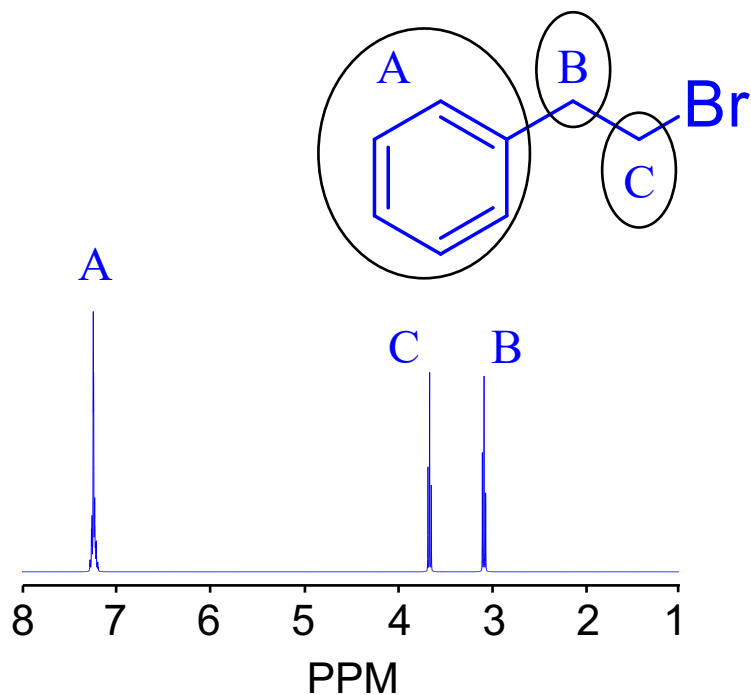
- 4-Bond (4J) Coupling

Video 5: Diastereotopic Protons

- 2-Bond (2J) Coupling

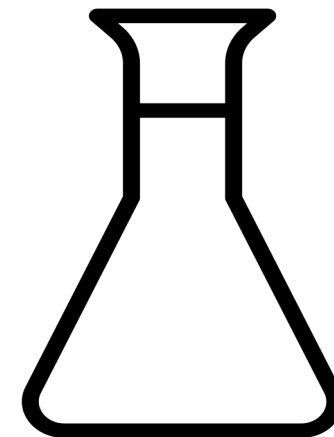
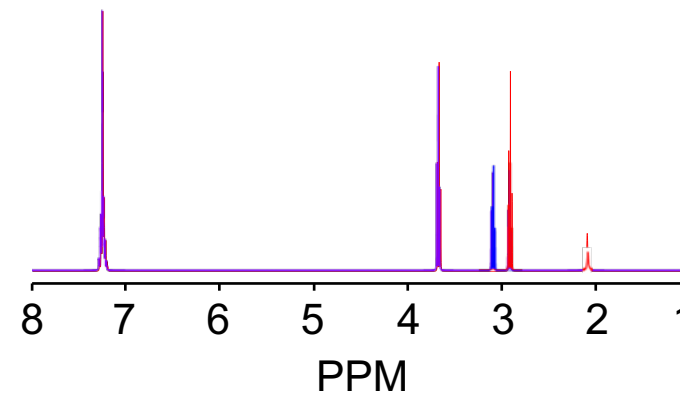
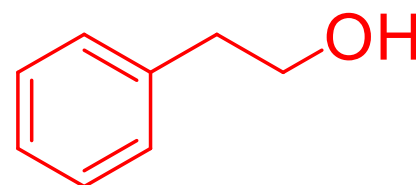
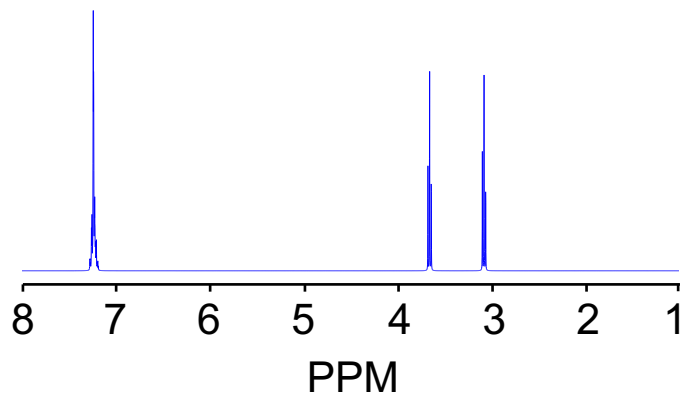
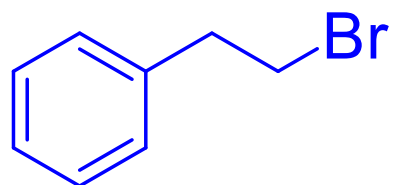
^1H NMR of Compound Mixtures

NMR data for two similar compounds



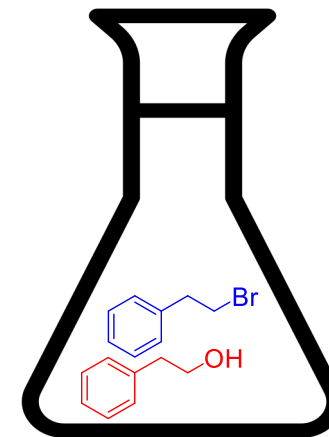
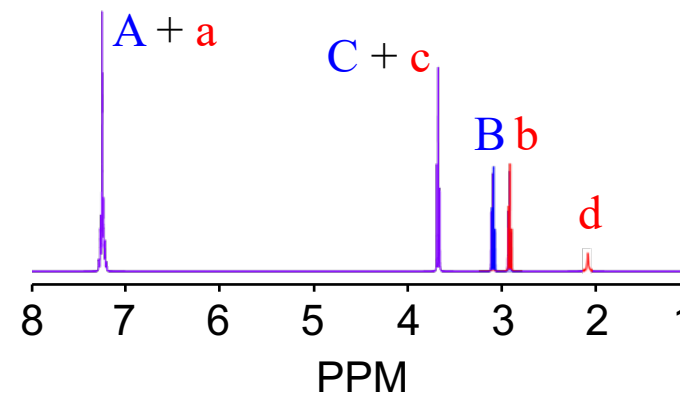
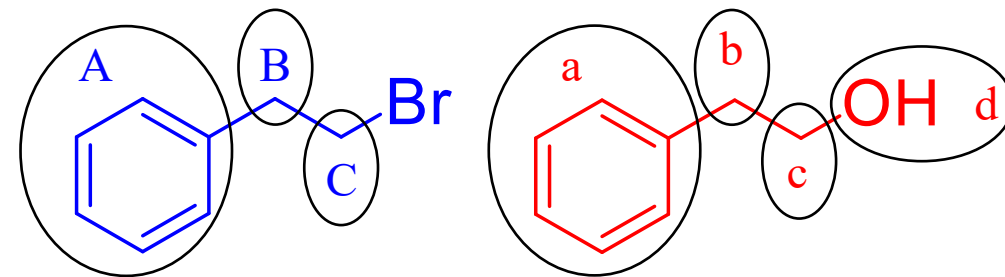
^1H NMR of Compound Mixtures

NMR data for a mixture of two compounds will have all signals for both compounds



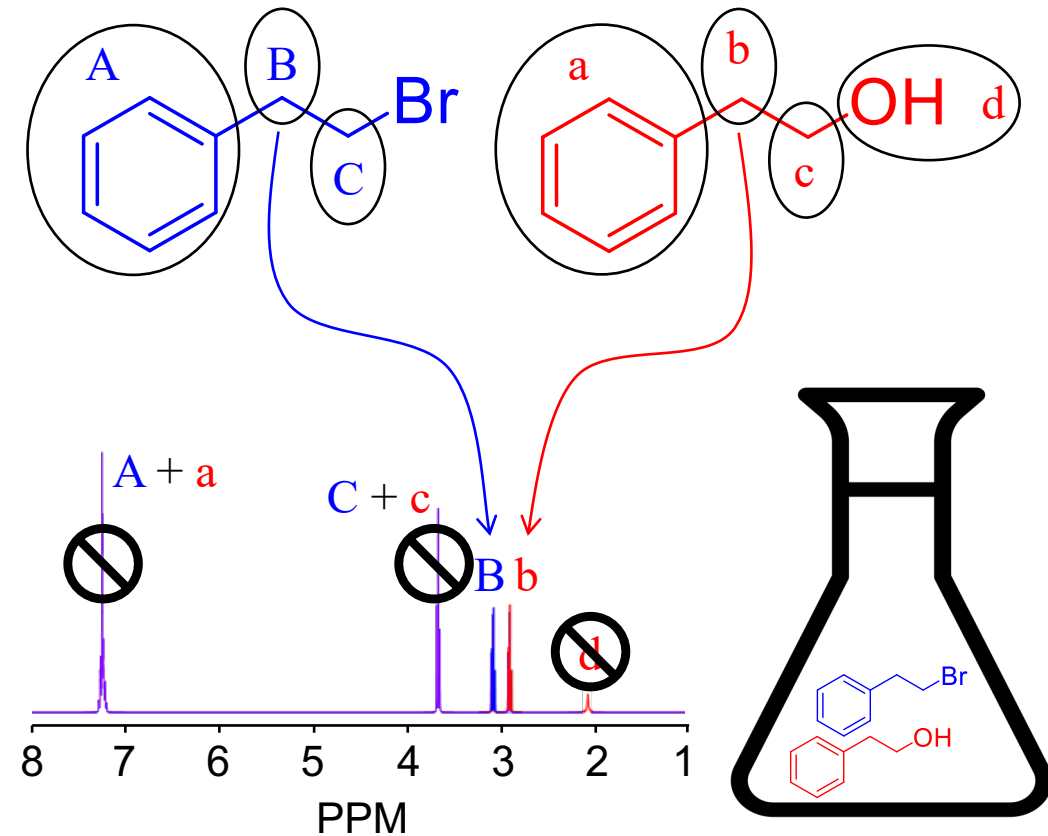
Diagnostic Signals

- Diagnostic signals uniquely identify a specific compound
- Integrations of diagnostic signals can be used to calculate the percent composition of a mixture.

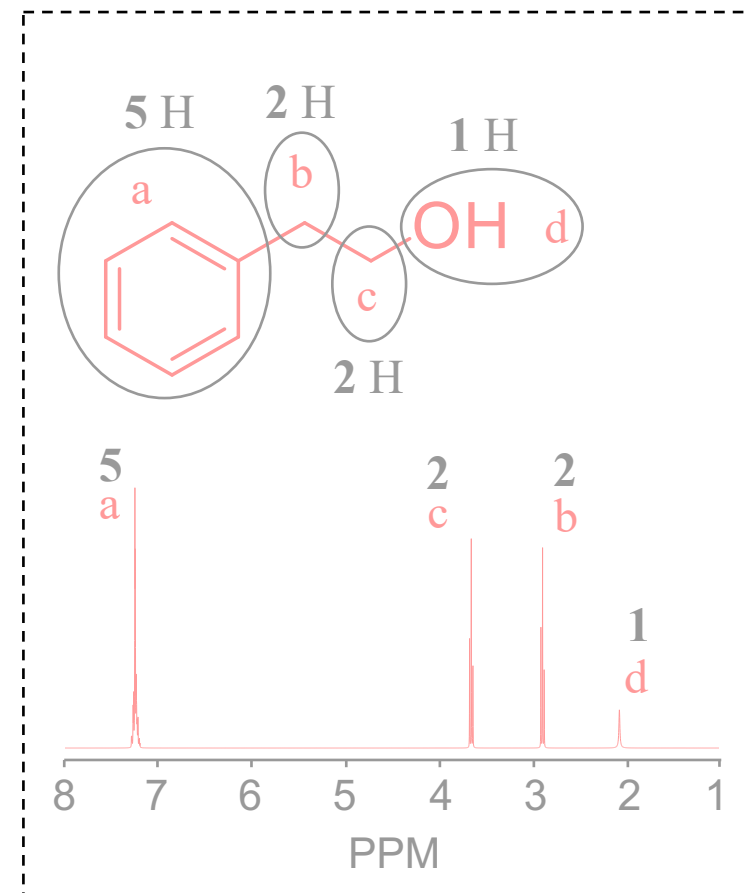
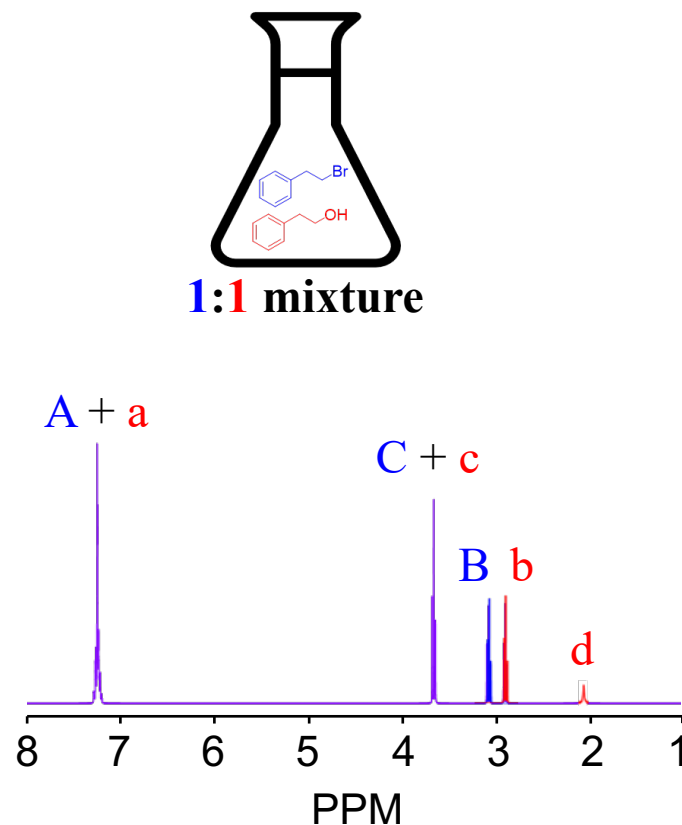
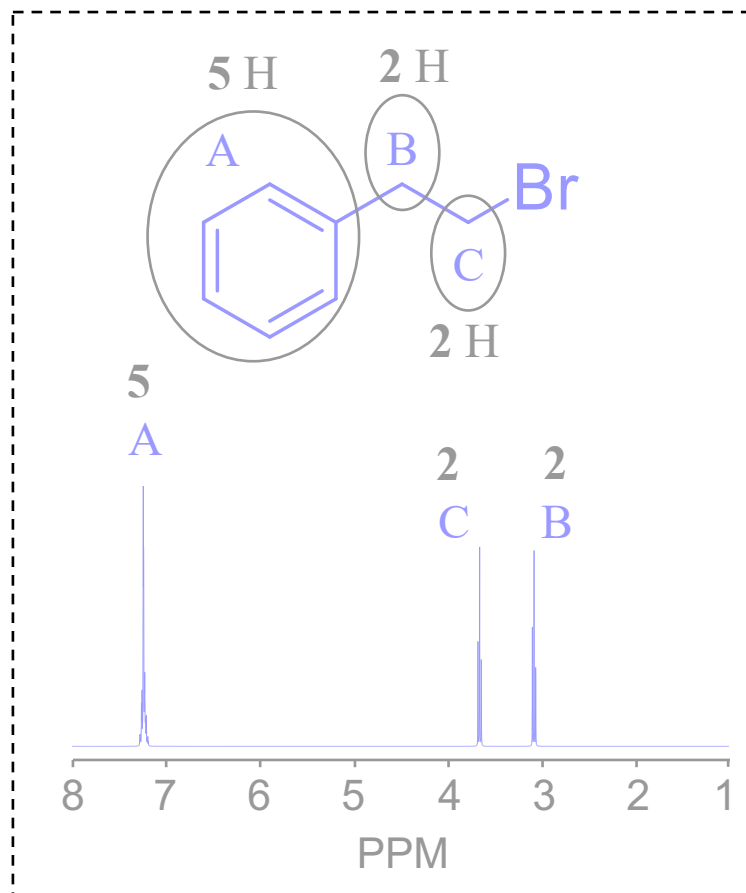


Diagnostic Signals

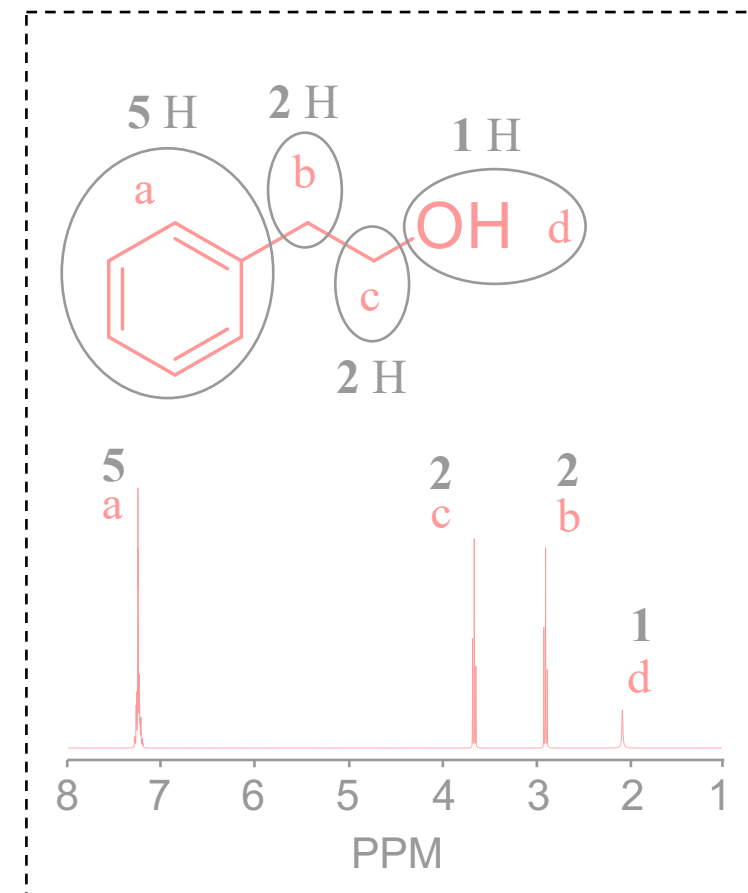
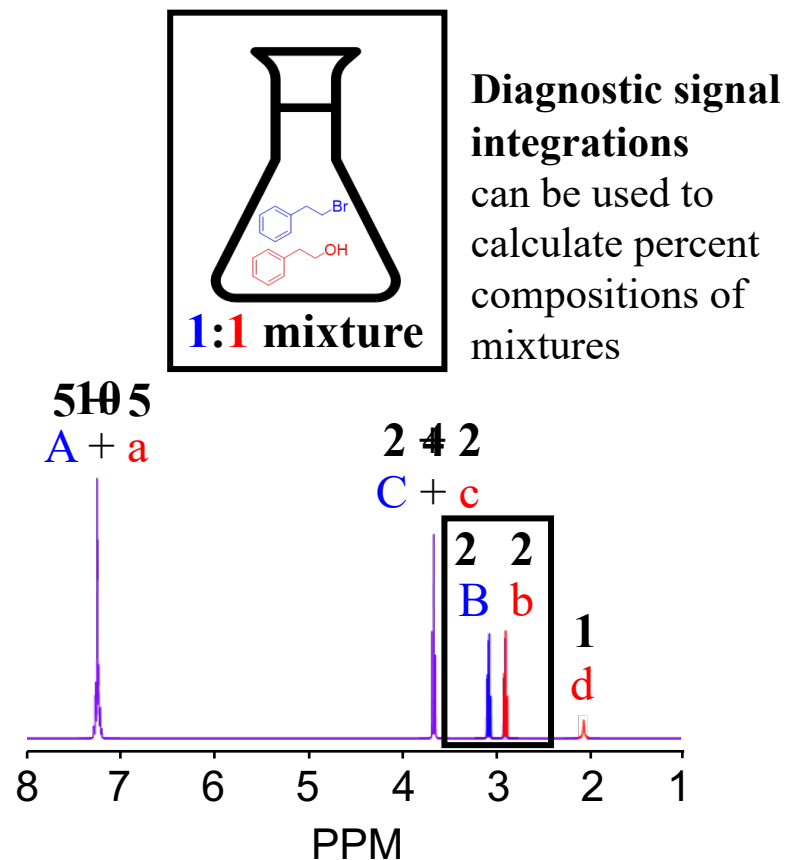
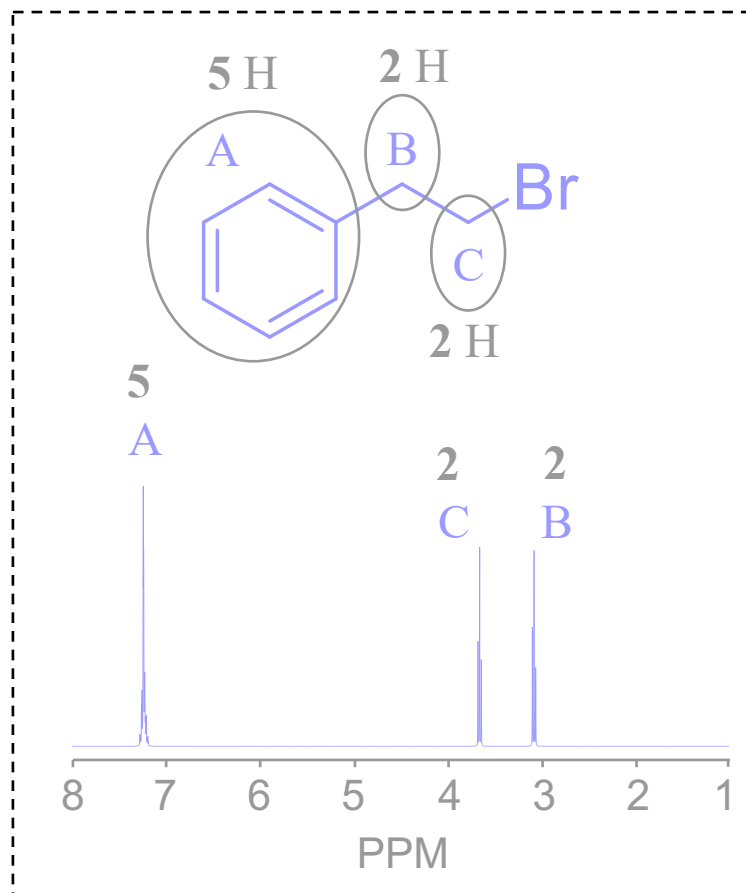
- **Diagnostic signals uniquely identify a specific compound**
- **Integrations of diagnostic signals can be used to calculate the percent composition of a mixture.**
- Must not overlap with signals from other compounds
- Avoid signals of exchangeable protons if possible (O-H, N-H)
 - These signals tend to have variable chemical shifts and do not always integrate reliably
- **Must be able to assign the signals to specific protons**
 - Knowing the number of protons that give rise to each signal is important!



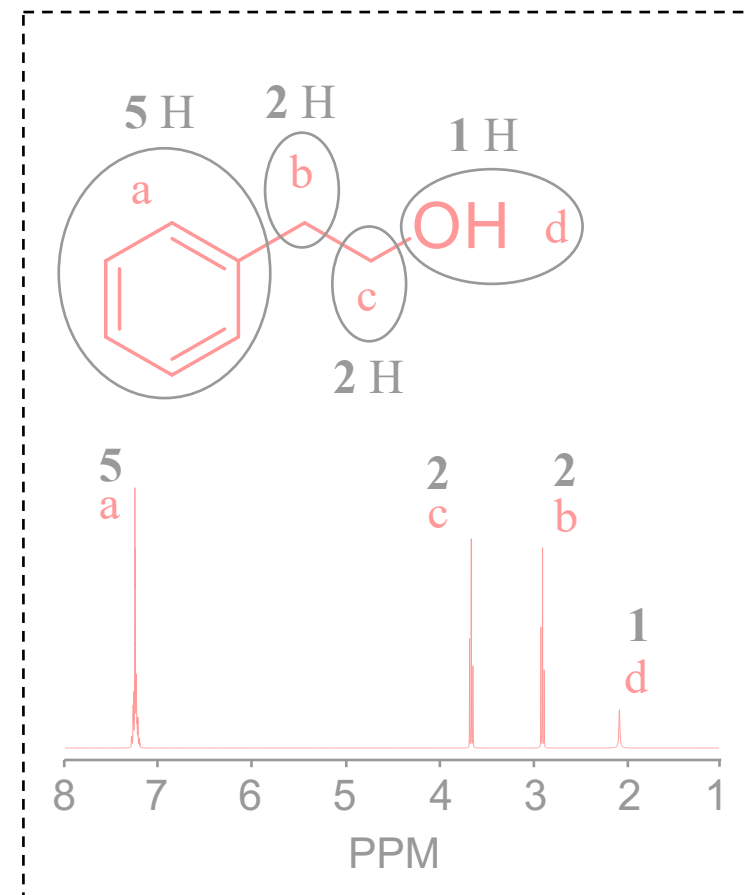
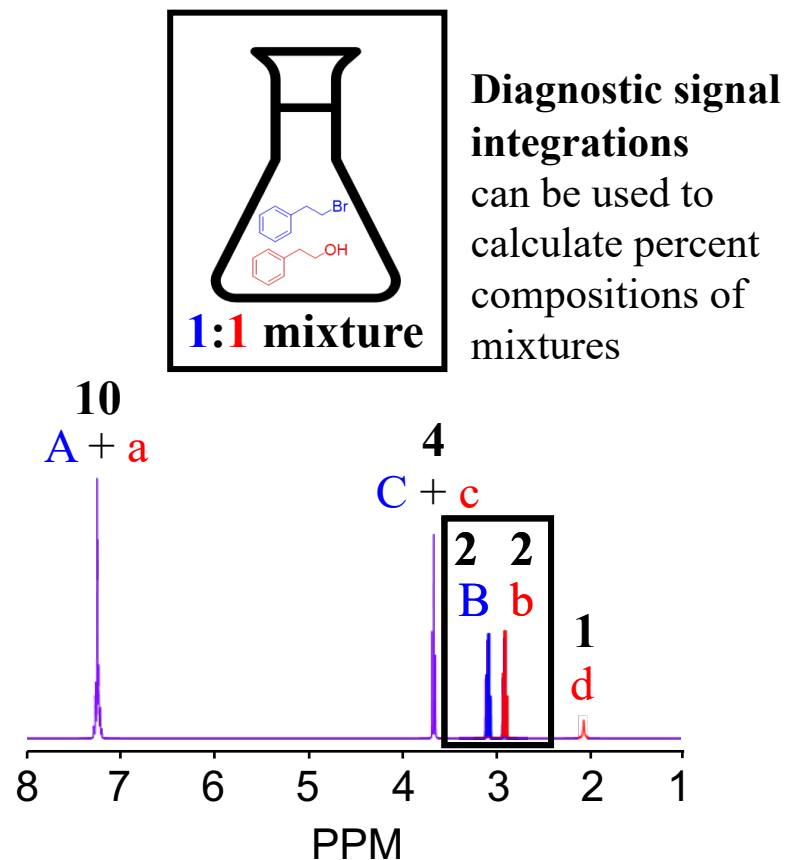
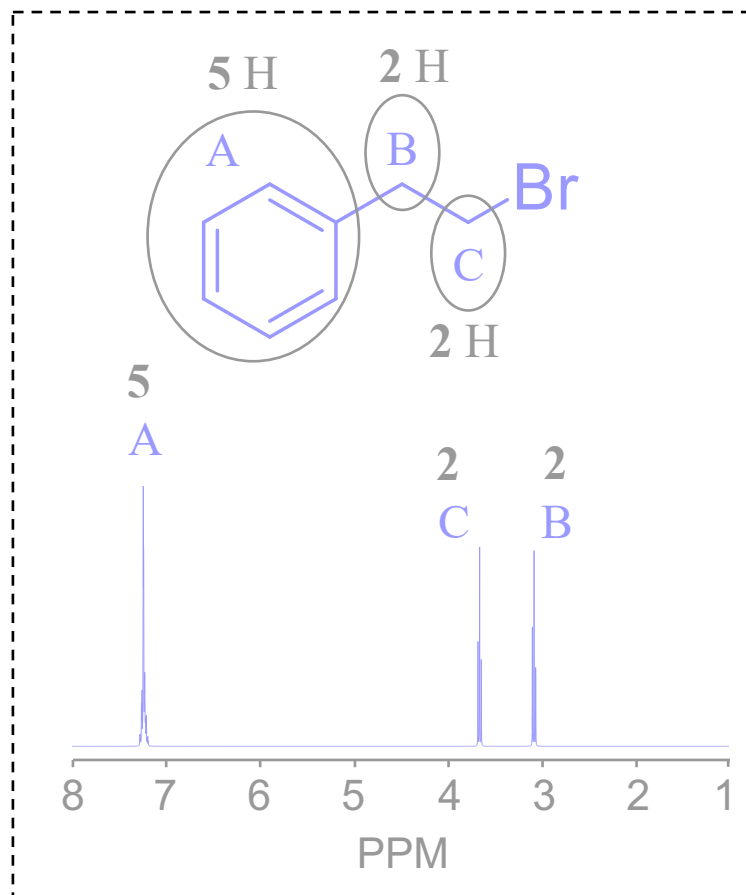
Signal Integrations of Compound Mixtures



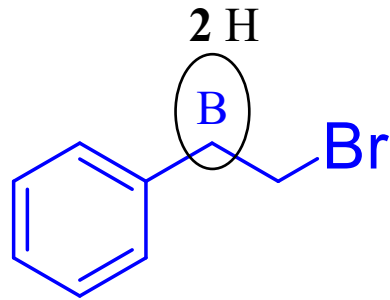
Signal Integrations of Compound Mixtures



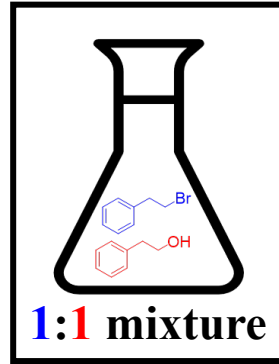
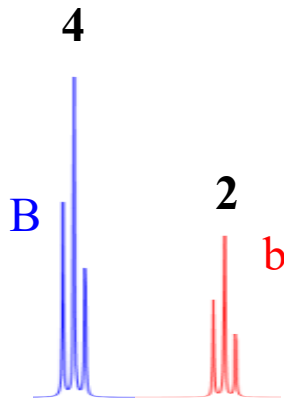
Signal Integrations of Compound Mixtures



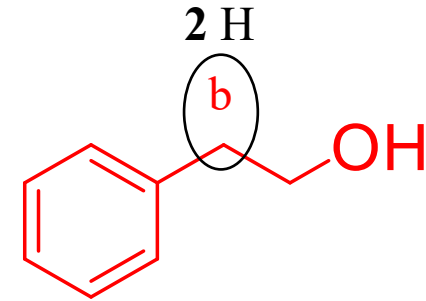
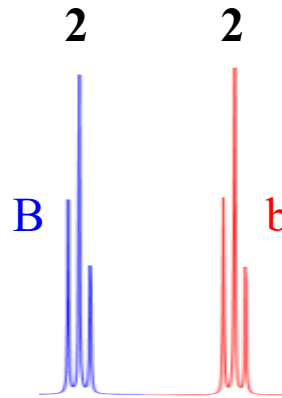
Signal Integrations of Compound Mixtures



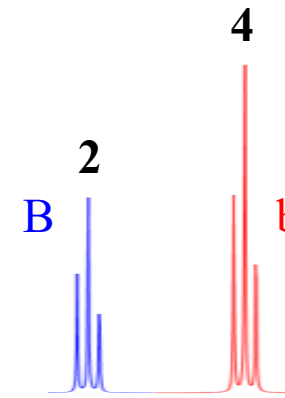
2:1 mixture



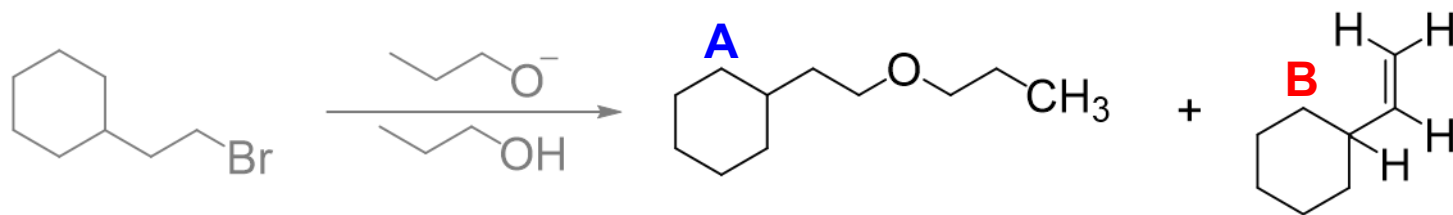
Diagnostic signal integrations can be used to calculate percent compositions of mixtures



1:2 mixture

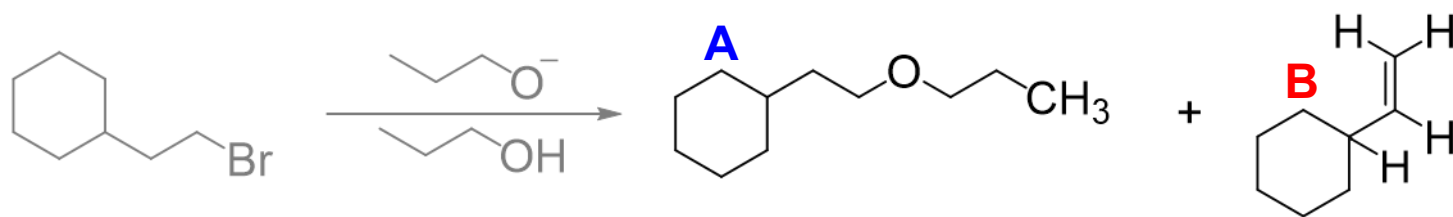


Calculating Percent Composition

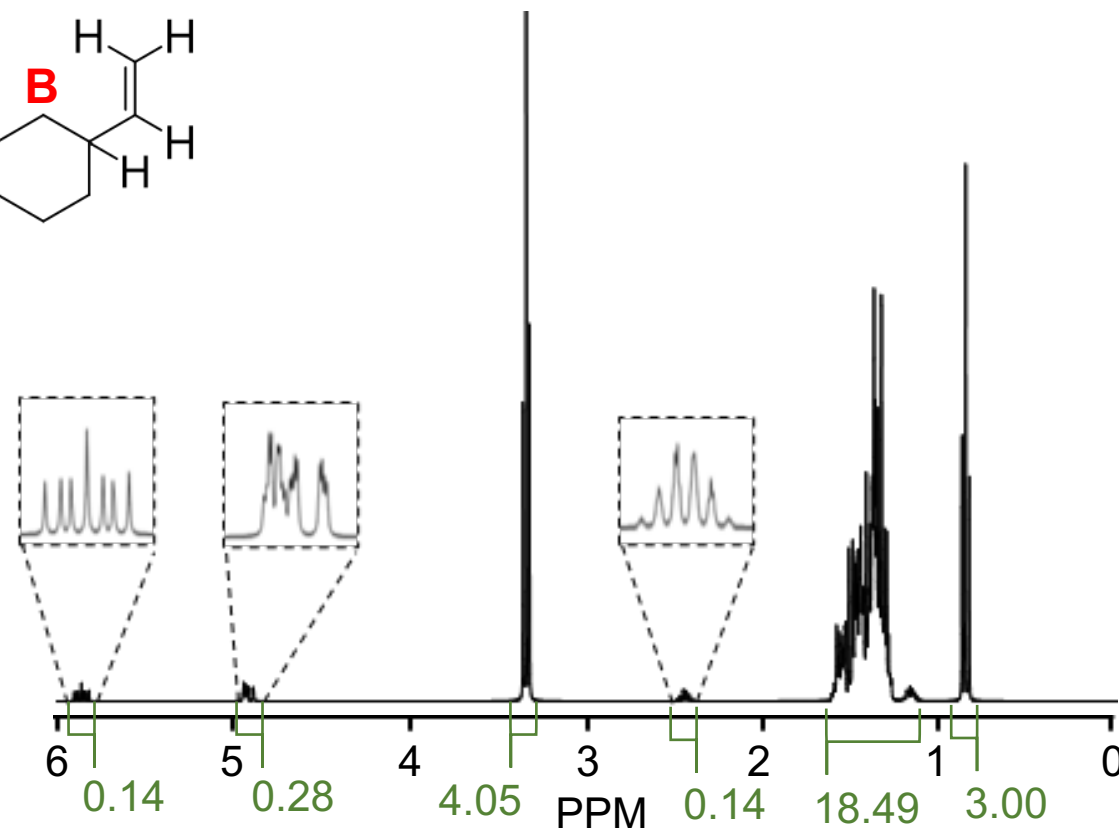


What is the mole ratio of products **A** and **B**?

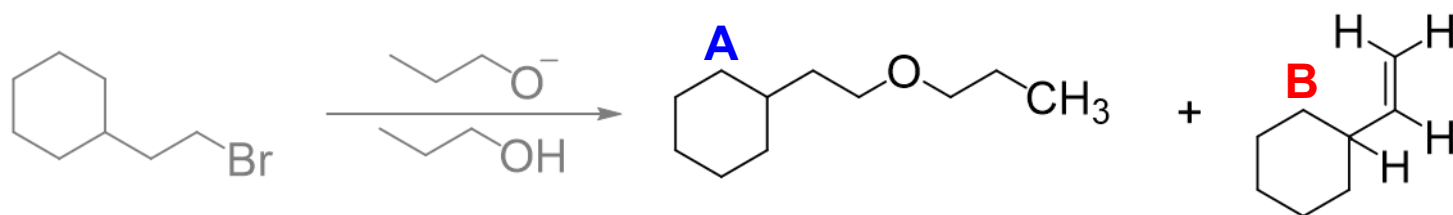
Calculating Percent Composition



What is the mole ratio of products **A** and **B**?

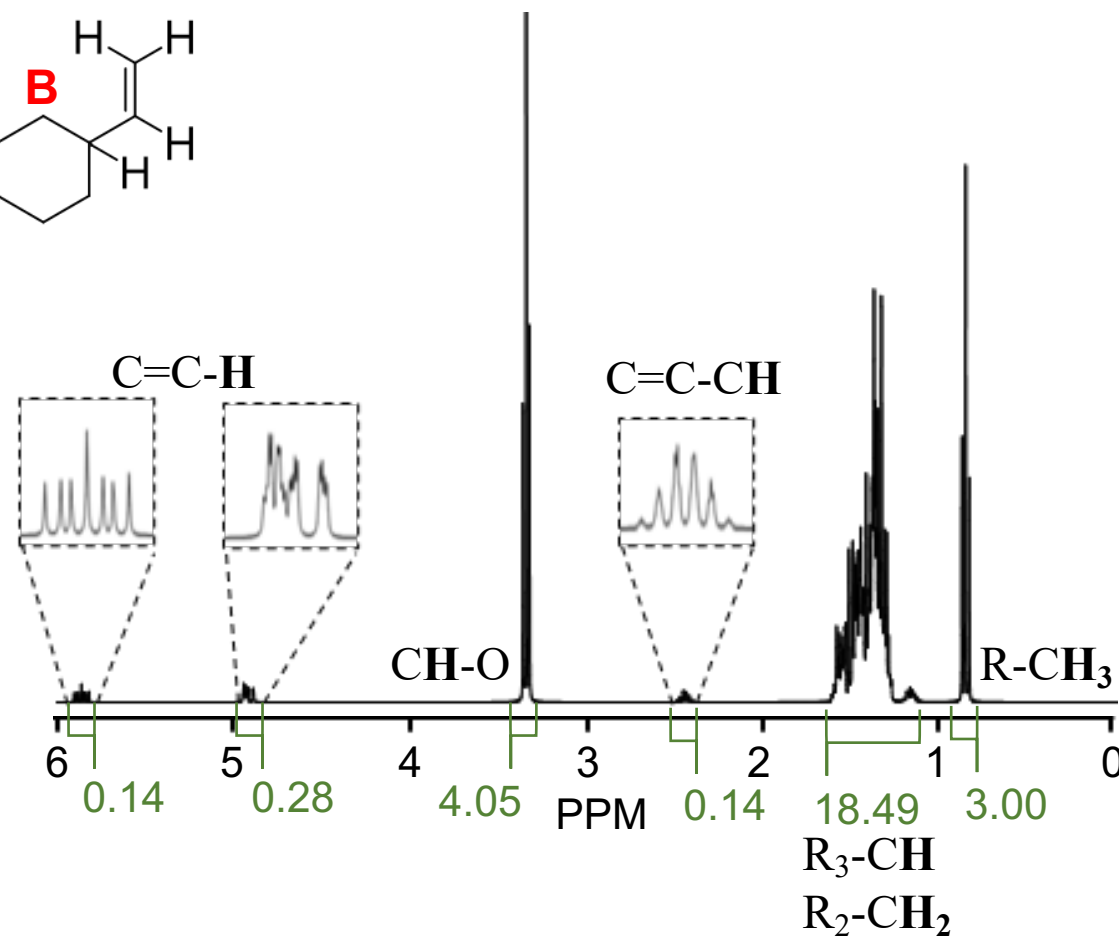


Calculating Percent Composition

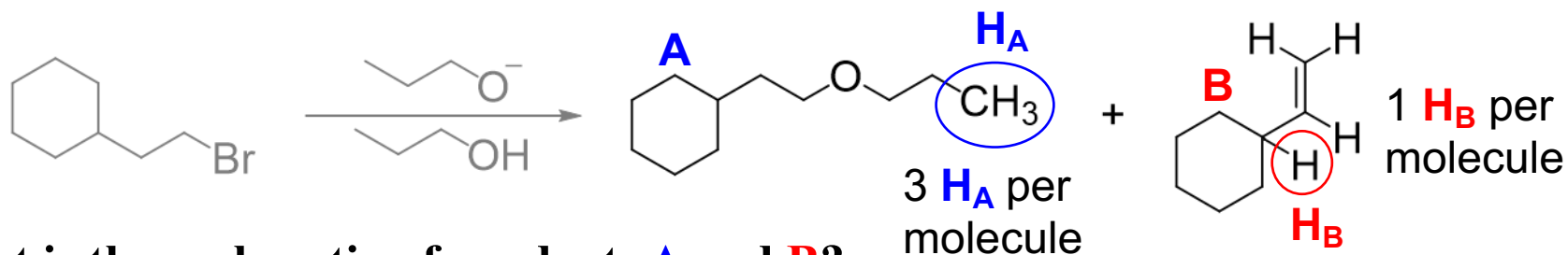


What is the mole ratio of products **A** and **B**?

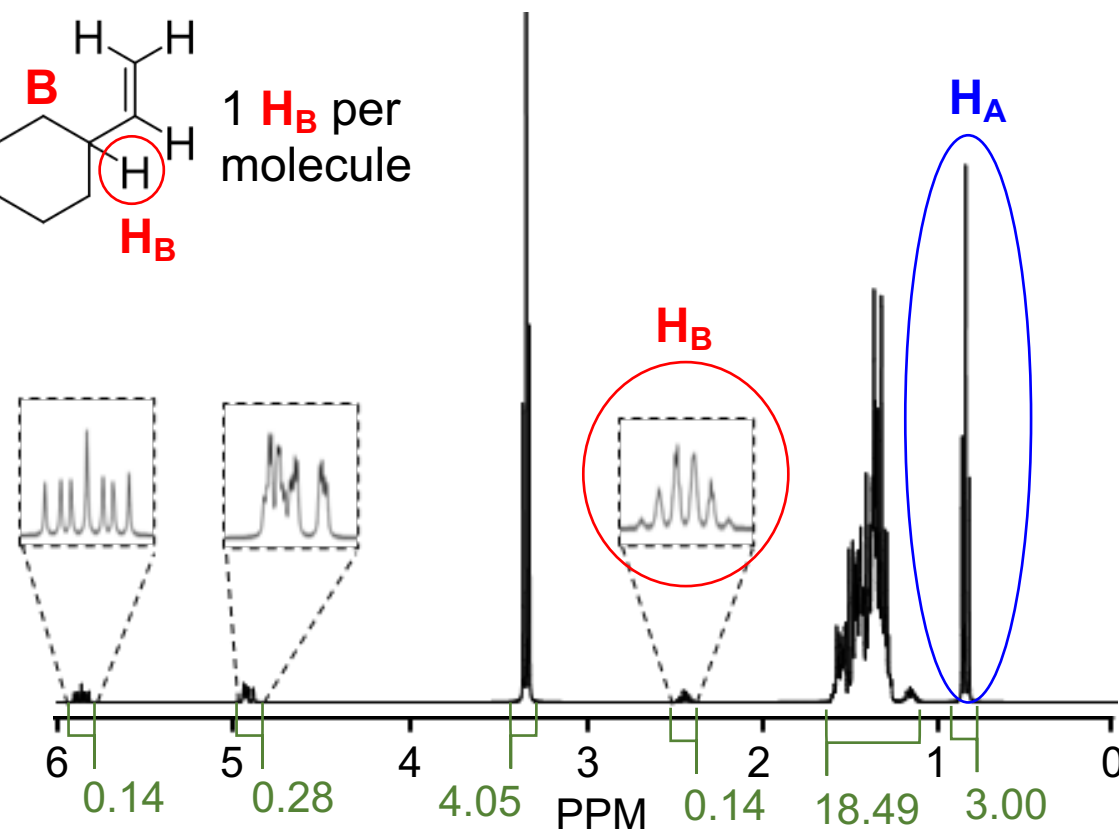
C=C-H	Alkenyl (Vinyl)	4.5 – 6
CH-O	Alcohol or Ether	3.3 – 4
R-CH ₃	Primary alkyl	~ 0.9
R ₂ -CH ₂	Secondary alkyl	~ 1.3
R ₃ -CH	Tertiary alkyl	~ 1.5
C=C-CH	Allylic	~ 1.7



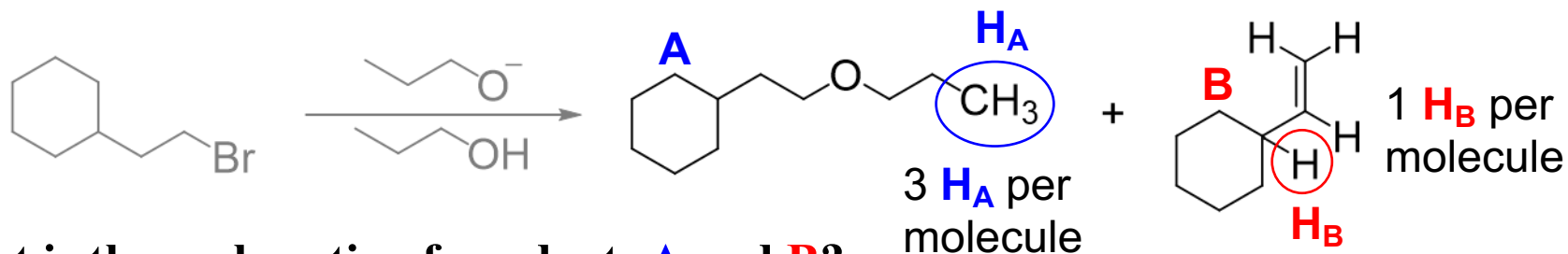
Calculating Percent Composition



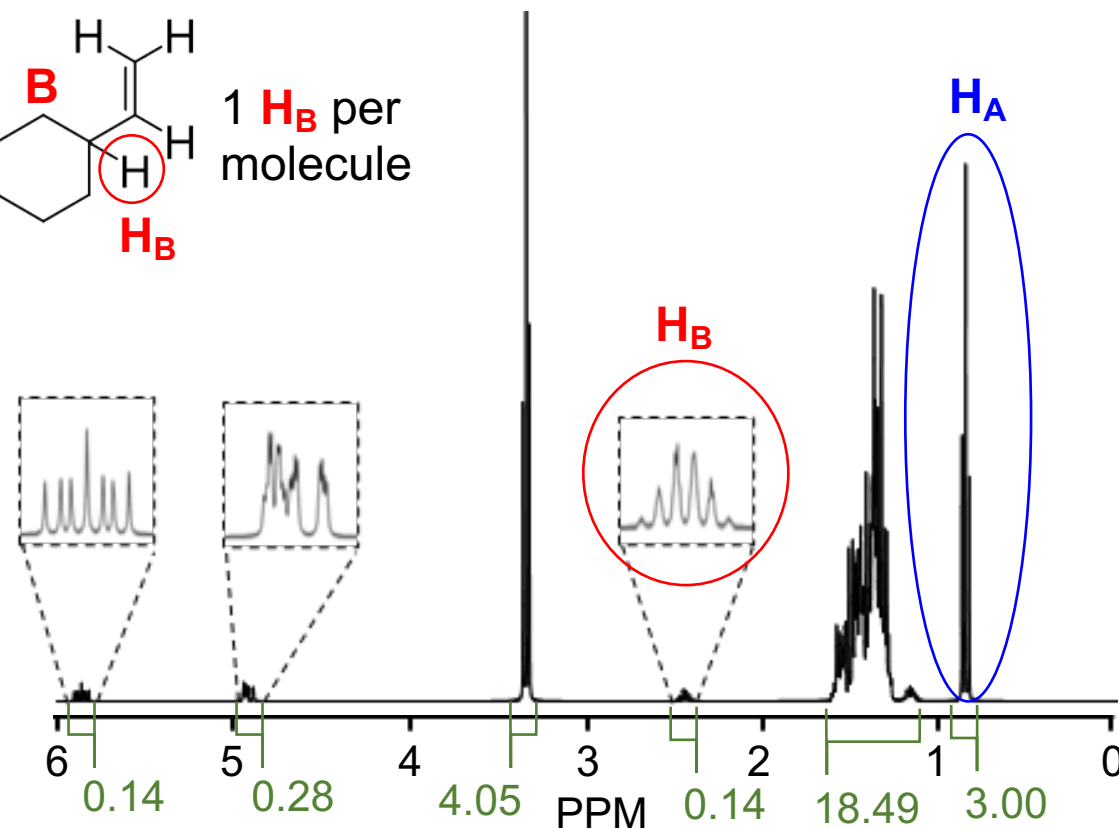
What is the mole ratio of products A and B?



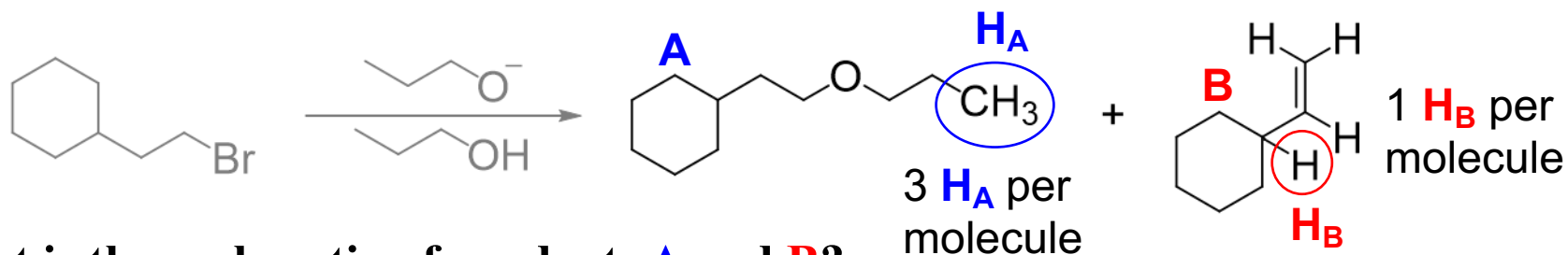
Calculating Percent Composition



What is the mole ratio of products **A** and **B**?



Calculating Percent Composition



What is the mole ratio of products A and B?

H_A integration: 3.00

Number of H_A protons: 3H

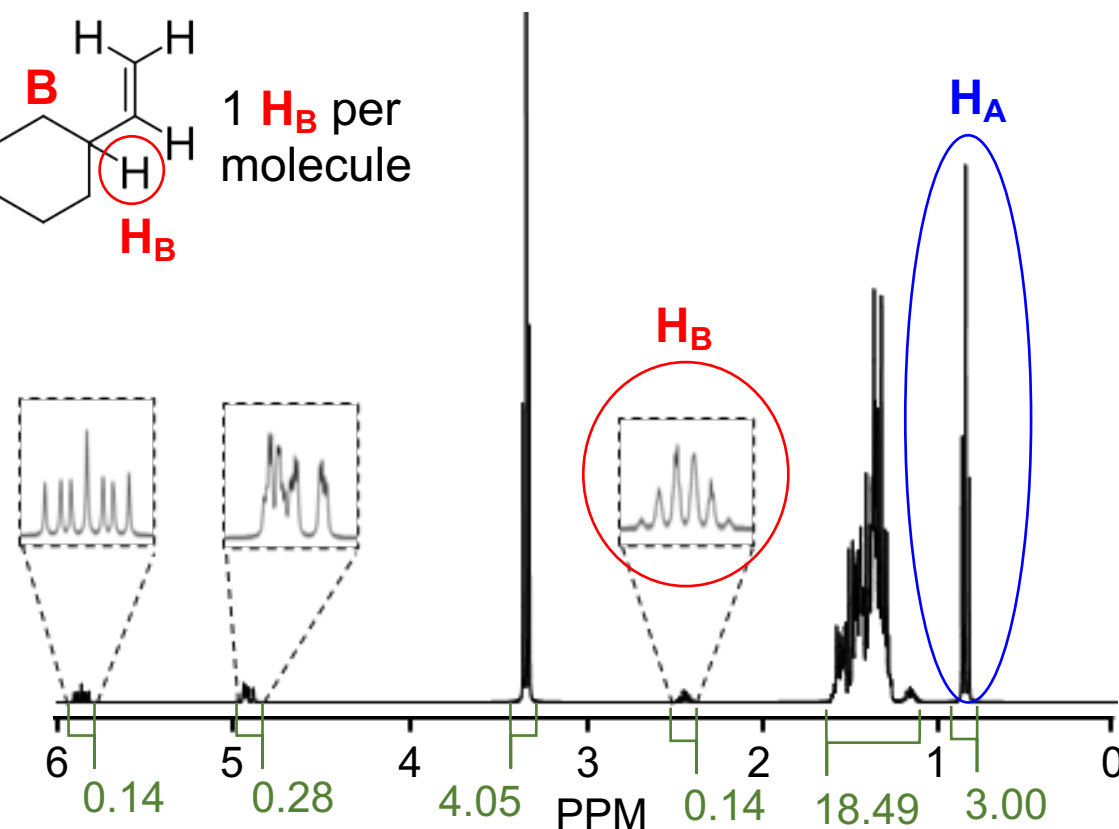
$$\frac{\text{Signal Integration}}{\text{Number of H per signal}} = \frac{3.00}{3H} = 1$$

H_B integration: 0.14

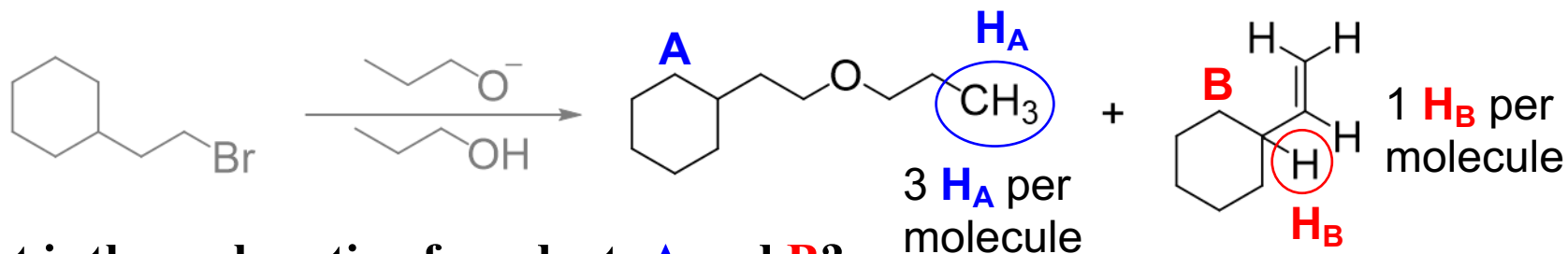
Number of H_B protons: 1H

$$\frac{\text{Signal Integration}}{\text{Number of H per signal}} = \frac{0.14}{1H} = 0.14$$

Ratio of
products
A and B
1 : 0.14



Calculating Percent Composition



What is the mole ratio of products **A** and **B**?

H_A integration: 3.00

Number of H_A protons: 3H

$$\frac{\text{Signal Integration}}{\text{Number of H per signal}} = \frac{3.00}{3H} = 1$$

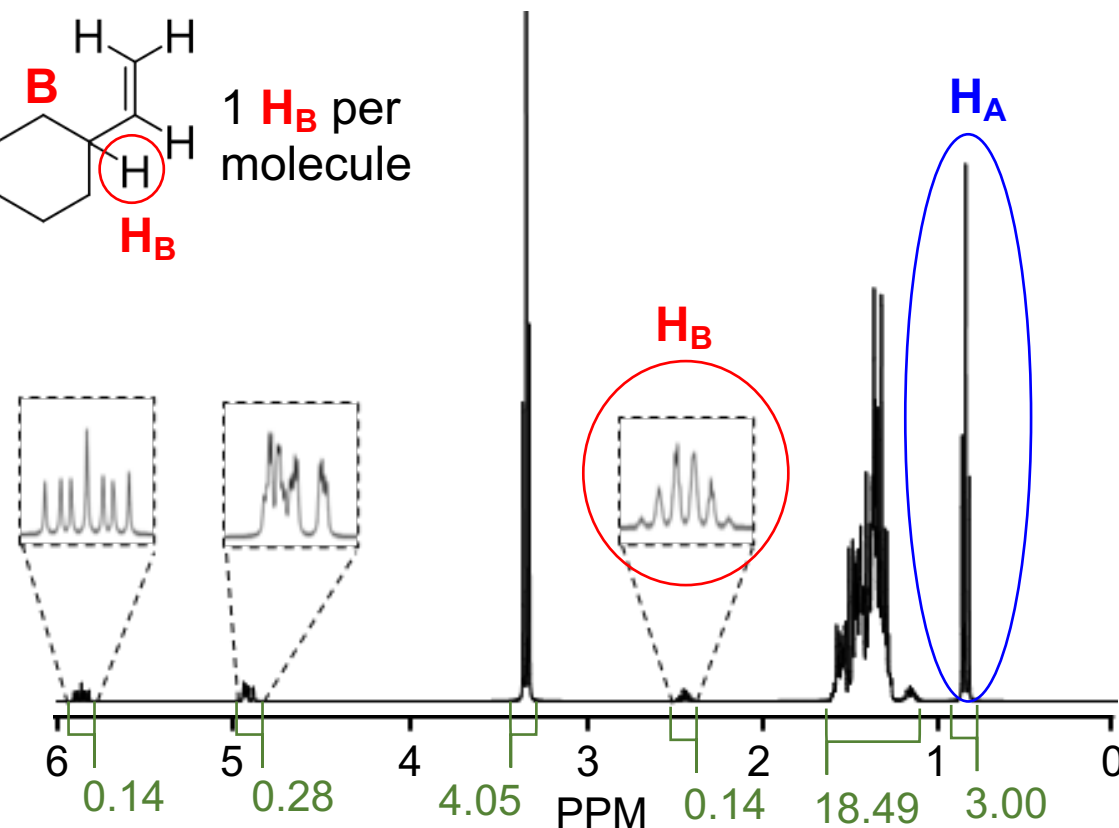
1 : 0.14

Ratio of
products
A and **B**
1 : 0.14

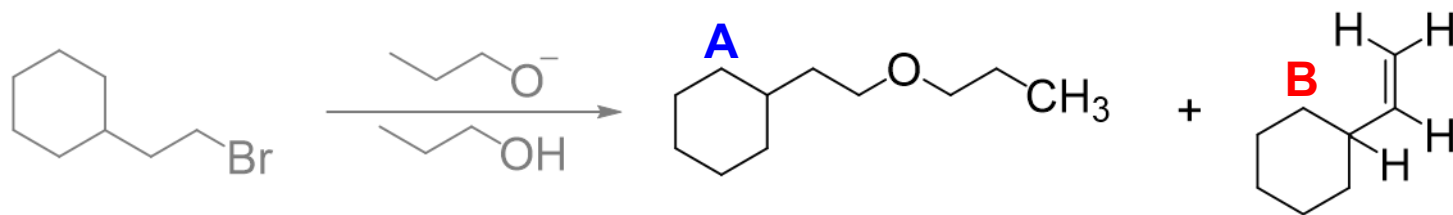
H_B integration: 0.14

Number of H_B protons: 1H

$$\frac{\text{Signal Integration}}{\text{Number of H per signal}} = \frac{0.14}{1H} = 0.14$$



Calculating Percent Composition



What is the mole ratio of products **A** and **B**?

1 : 0.14

What is the percent composition of **A**?

$$\% \text{ Composition of } \mathbf{A} = \left(\frac{\text{Integration due to } \mathbf{A}}{\text{Total Integration } (\mathbf{A}+\mathbf{B})} \right) * 100 = \left(\frac{1}{1+0.14} \right) * 100 = 87.7\% \mathbf{A}$$

$$\% \text{ Composition of } \mathbf{B} = \left(\frac{\text{Integration due to } \mathbf{B}}{\text{Total Integration } (\mathbf{A}+\mathbf{B})} \right) * 100 = \left(\frac{0.14}{1+0.14} \right) * 100 = 12.3\% \mathbf{B}$$



Video Summary

Diagnostic NMR signals uniquely identify a compound

Use diagnostic signals to calculate the percent composition of a mixture

- Avoid using exchangeable protons (O-H, N-H) if possible
- Assign diagnostic signals to specific proton environments in each compound
- Divide the integration for each diagnostic signal by the number of protons that give rise to the signal

This step is needed to relate the integration value to the relative number of molecules

- **% Composition** = $\left(\frac{\text{Integration due to one compound}}{\text{Total (Integration due to all compounds)}} \right) * 100\%$

Next video: NMR Solvents